Monte Carlo Simulation of Neutron Slowing Down

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Based on Wigner–Wilkins (1944), R. R. Coveyou, R. R. Bate, and R. K. Osborn (1955)

Nuclear Fission

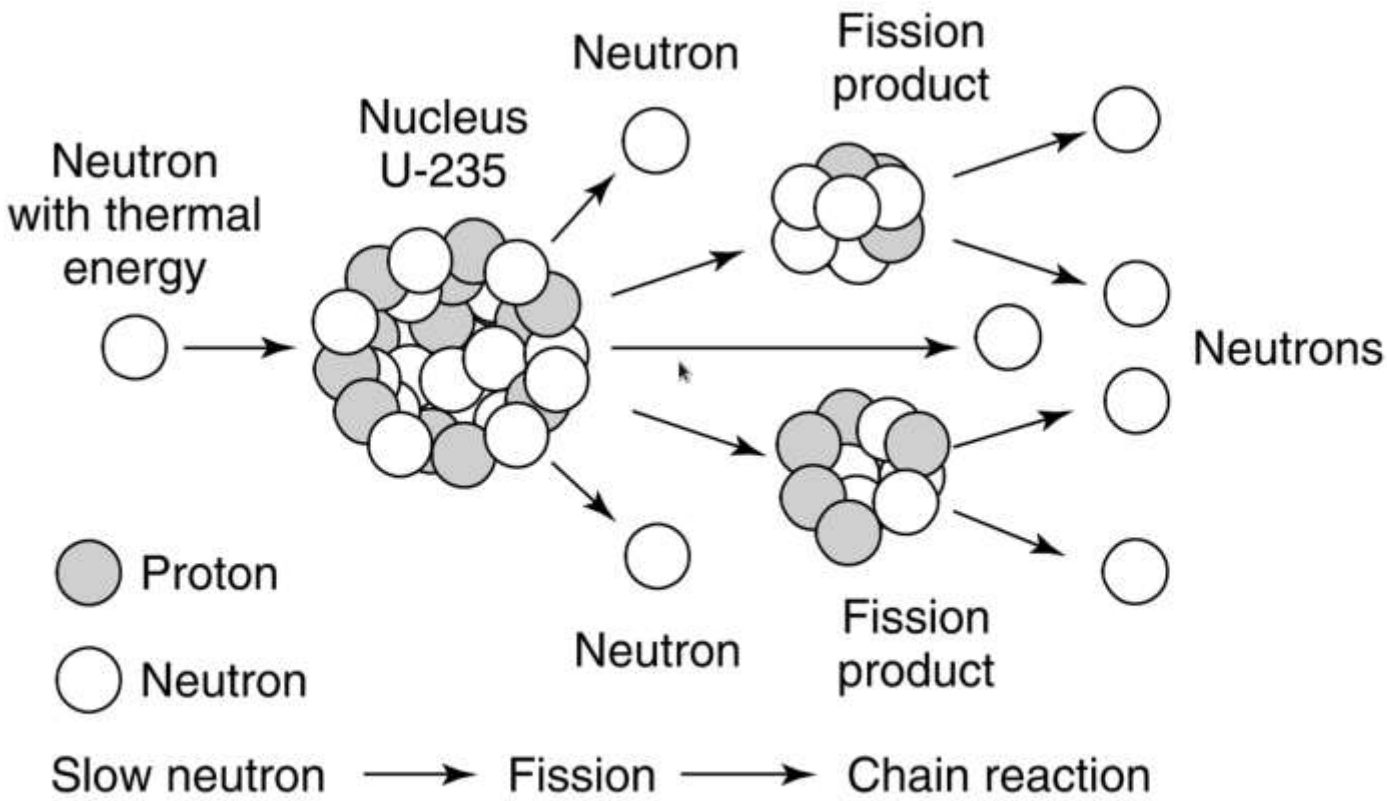


Fig. 1.10 Schematics of fission processes. Filges, D. & Goldenbaum, *Handbook of Spallation Research*, Wiley, 2010

Neutron Moderator

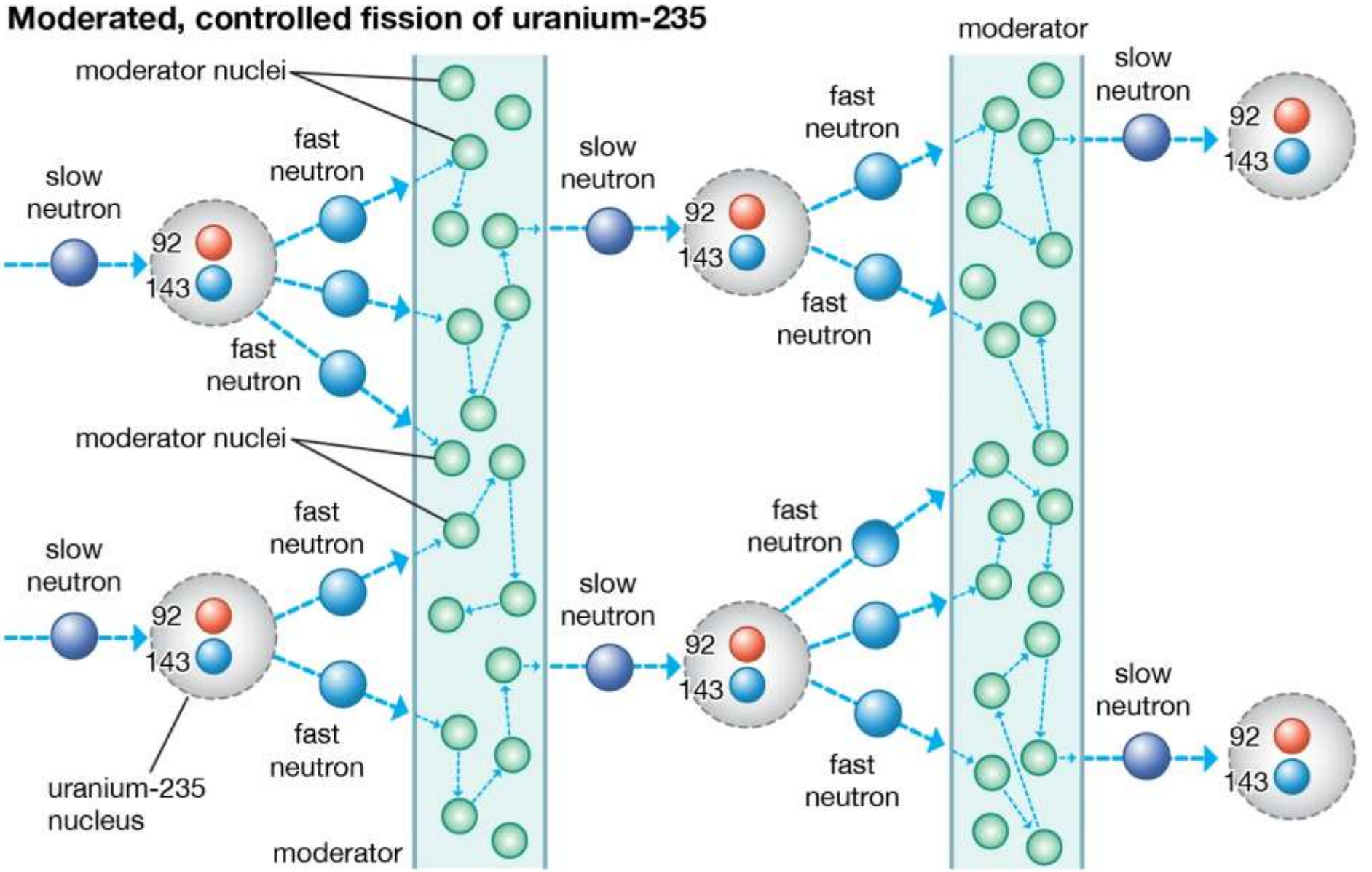
We want Chain Reaction

But primary neutrons are too fast
and cause low probability of being absorbed by the Uranium nucleus

We want to slow it down

Then we use Neutron Moderator

Common Moderators: H2O, D2O, Graphite, Be


Moderated, controlled fission of uranium-235
moderator nuclei
slow neutron
92
143
fast neutron
fast neutron
moderator nuclei
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uranium-235 nucleus
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Eugene Wigner & J. Ernest Wilkins

Effect of the Temperature of the Moderator on the Velocity Distribution of Neutrons with Numerical Calculations for H as Moderator (1944)

Thermal region: Neutrons accumulate near thermal energies, forming a Maxwellian-like peak.

High-energy region: Absorption suppresses the classical $1/v^2$ tail.

Limitation: Suitable only if the mass of moderator = neutron mass

(H Moderator)

R. R. Coveyou, R. R. Bate, and R. K. Osborn

(1955)

Oak Ridge National Laboratory

Effect of Moderator Temperature upon Neutron Flux in Infinite, Capturing Medium

- Validated Wigner & Wilkins for hydrogen
- Extended neutron moderation theory to all moderators
- Revealed real thermal spectral distortions

Why Monte Carlo?

Deterministic analytical equations extremely complex

Random velocities, angles, and absorption probabilities

Monte Carlo naturally models probabilistic behavior

Original Monte Carlo neutron studies were by Los Alamos team

The Code - Overview

Written in C

Recreates Monte Carlo method from the paper

Implements all physical effects:

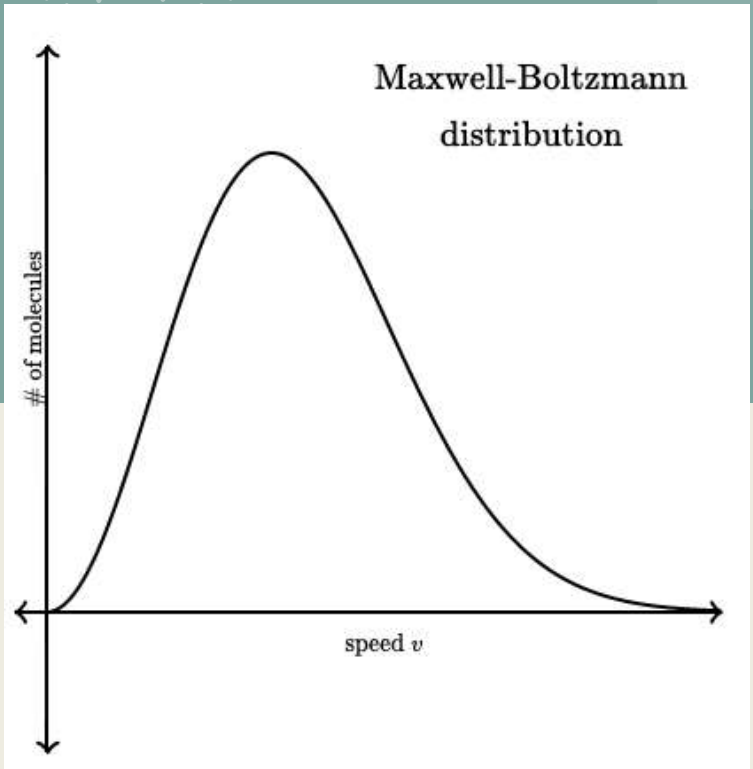
- Maxwell–Boltzmann target sampling
- Effective scattering cross-section
- 1/v absorption
- Elastic CM collision physics

Tallies neutron speeds → flux spectrum

Code Structure

```
main()
 ├── init_simulation()          ← Set constants, moderator A, absorption K, bins
 ├── run_simulation()           ← Loop over 1,000,000 neutron histories
 │    └── run_history()         ← One neutron's life: scatter or absorb
 │         ├── calculate_effective_sigma_s()
 │         ├── sigma_a(v) = σa0 / v
 │         ├── sample_target_velocity()
 │         ├── elastic_collision()
 │         └── tally speed into bins
 └── calculate_flux()       ← Convert tallies → φ(v)
```

Sampling Target Velocities



Maxwell–Boltzmann distributed speeds

Target velocity **must** be sampled with bias

= relative speed

Use **rejection sampling** with acceptance probability:

$$P = \frac{v_{rel}}{v + V}$$

Exactly the "Kahn method" from the paper

Collision Physics

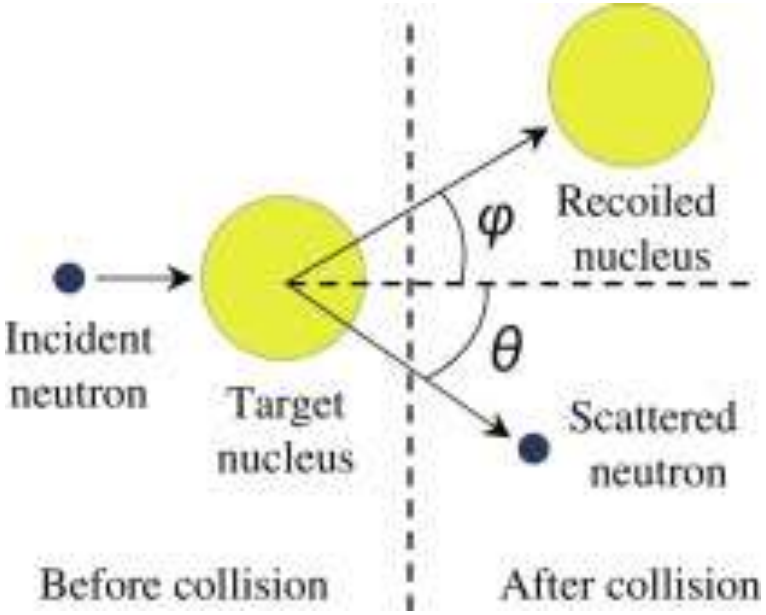
Compute center-of-mass velocity

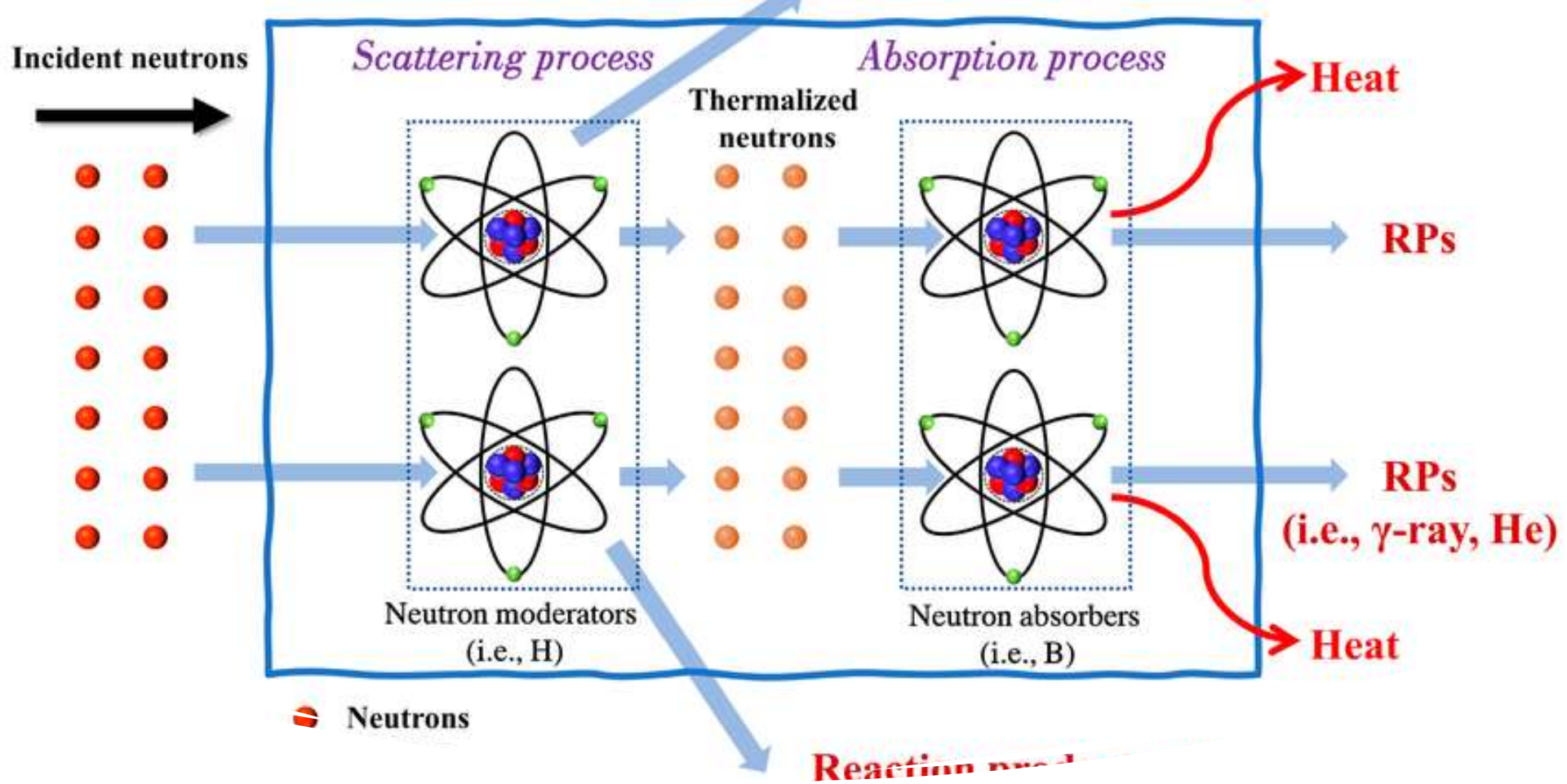
Sample isotropic angle in CM

Convert back to lab frame

Gives new neutron speed after elastic scattering

Momentum and energy conserved





Absorption vs. Scattering

Absorption follows 1/v law: $\sigma_a(v) = \frac{\sigma_{a0}}{v}$

Compute probability: $P_a = \frac{\sigma_a}{\sigma_a + \sigma_{s,eff}}$

Randomly absorb or scatter

Absorption ends the history

Flux Reconstruction

- Tally scattering events in speed bin iii
- Convert tallies ⟶ flux:

$$\Phi(v_i) \propto \frac{N_{scat}(v_i)}{N_{hist}\sigma_{s,eff}(v_i)\Delta v}$$

Normalize: peak thermal region = 1

- Export CSV for plotting

Results: Hydrogen

Peak at y ≈ 0.84

Good agreement with Wigner–Wilkins

1/v tail emerges at high speed

Minimal distortion of thermal peak

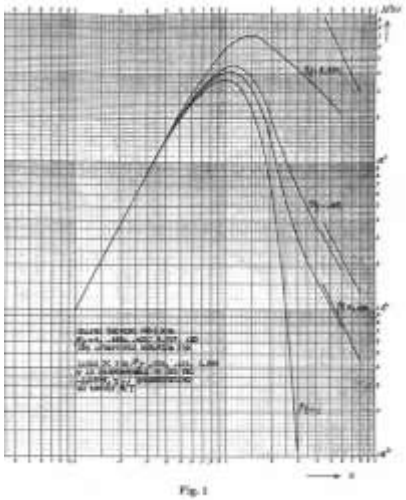


Fig. 1

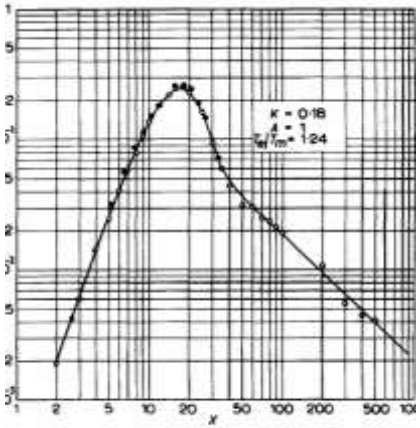
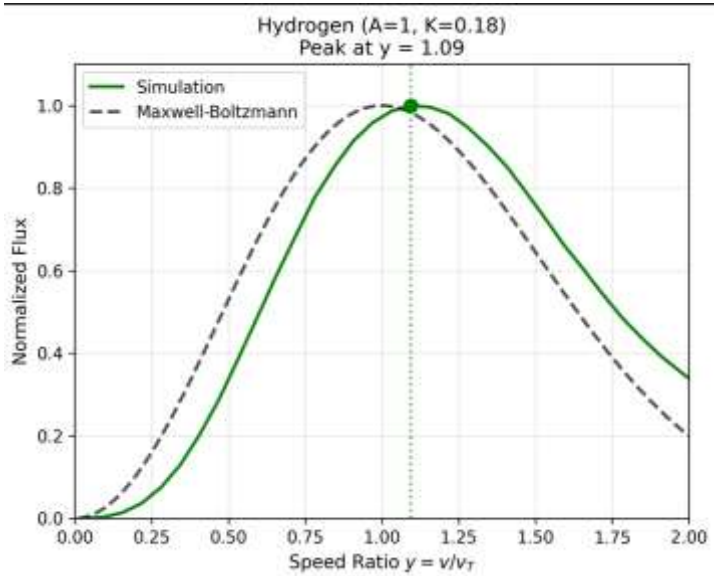


FIG. 12.
○ actual flux computed by Monte Carlo method
• actual flux computed by WIGNER and WILKINS



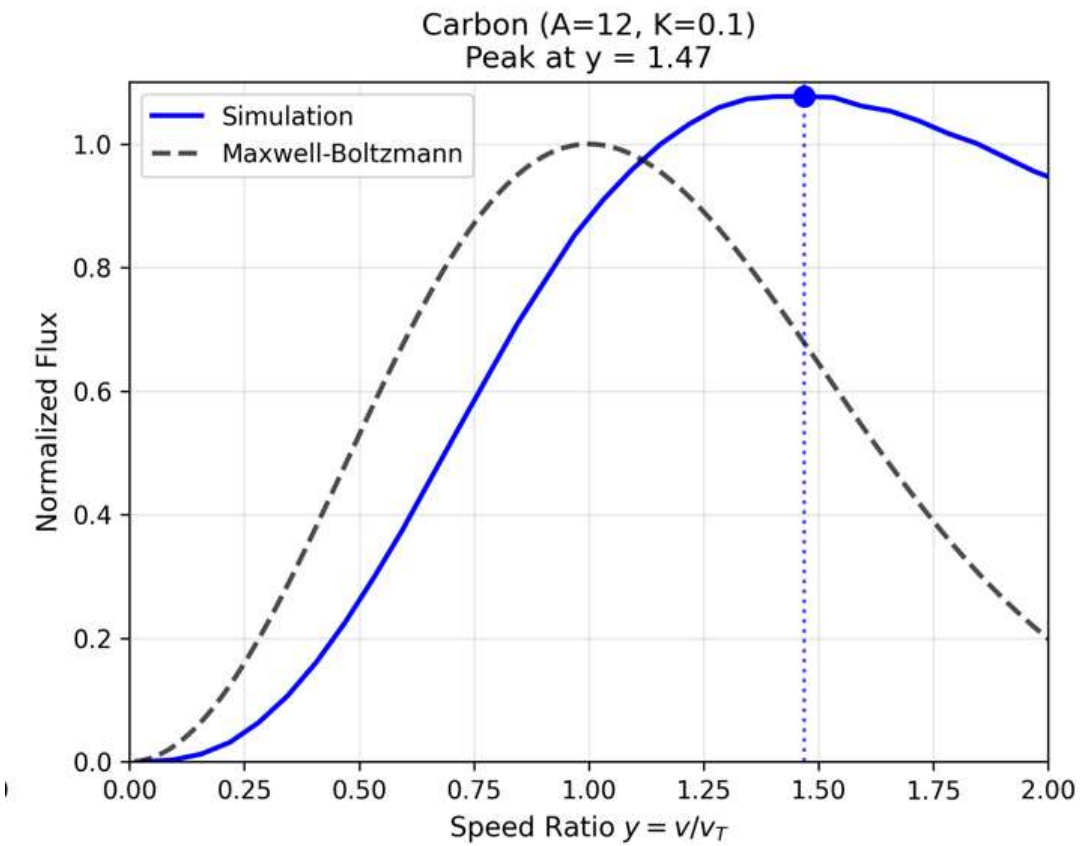
Results: Carbon

Peak at y ≈ 0.47

Heavier moderator ⟶
less energy loss per collision

Strong spectral hardening

High-energy tail elevated



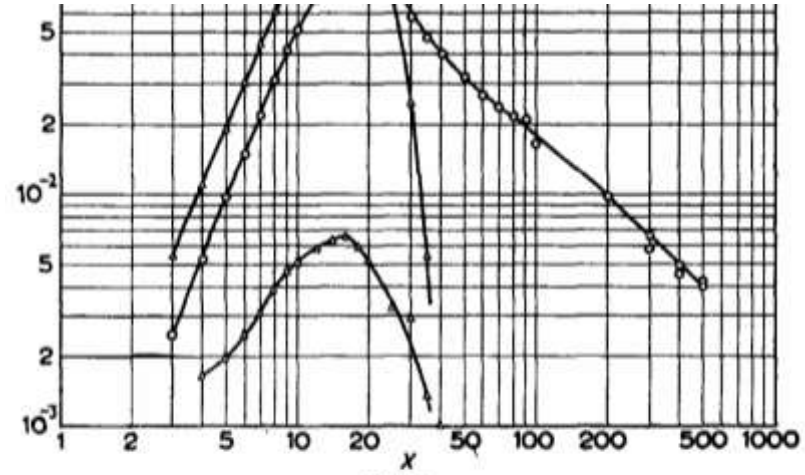
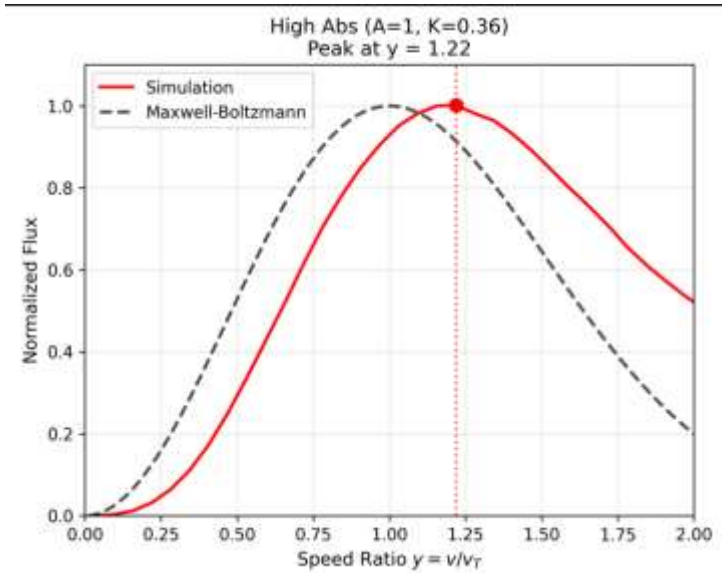


FIG. 1.

○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X
□ $(1/v)$ flux with same normalization



Results:
High Absorption

Low-energy suppression

Peak shifts higher ($y \approx 0.91$)

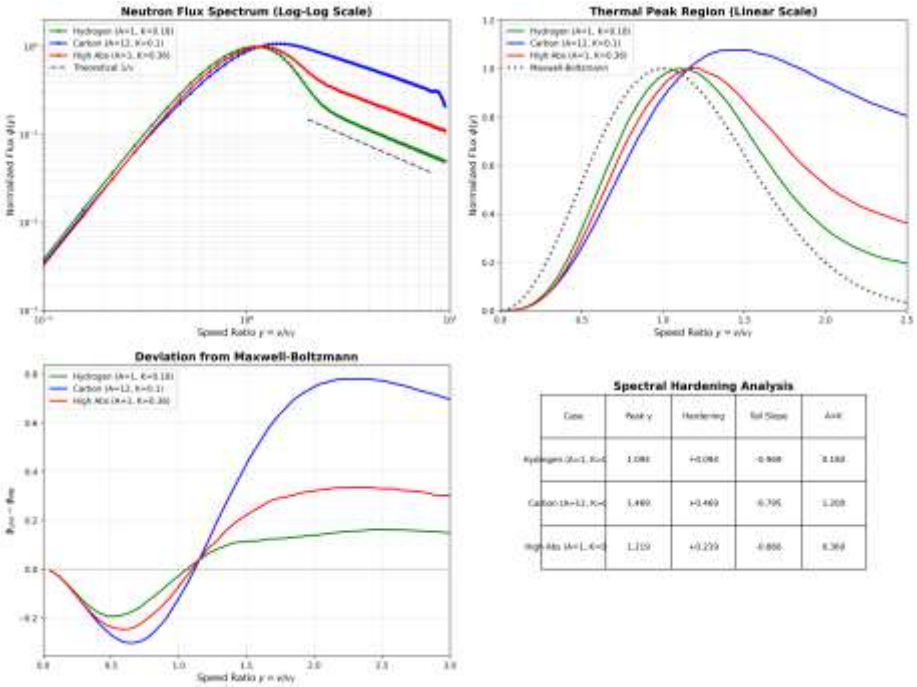
Distortion consistent with paper

Comparisons

All cases follow 1/v tail behavior

Deviation plots match published trends

Good reproduction of thermal distortions reported by the paper



Validation

From Coveyou et al.,

$$T_e/T_m \approx 1 + 1.11\ AK$$

within ~5 per cent

(Tm/Te=y_peak^2)

Moderator	A·K	Measured y_peak	Implied T_e/T_m	Predicted T_e/T_m=1+1.11A K
H (A=1,K=0.18)	0.180	1.094	1.196	1.1998
H (A=1,K=0.36)	0.360	1.218	1.483	1.3996
C (A=12,K=0.10)	1.200	1.469	2.157	2.3320

Limitations

- non-(1/v) absorbers, e.g. U-238
- mixtures of both moderators and absorbers
  Monte Carlo is still possible — but more complex

Acknowledgement
Geant4

- Developed by CERN
- for high energy physics, nuclear physics, etc
- Free access to public
- In C++

Conclusion

- Monte Carlo accurately models neutron slowing down in real moderators
- Results reproduce Wigner–Wilkins theory for hydrogen
- Results also match Oak Ridge Monte Carlo for heavier moderators and absorbers
- Simulation correctly shows thermal peak behavior, spectral hardening, and 1/v tails
- Code fully reconstructs the classical 1950s neutron moderation method
- Provides insight into how moderator mass and absorption shape the spectrum
- Establishes a foundation for more advanced, realistic reactor simulations

Thank You!